

4.13.105

AI25BTECH11036-SNEHAMRUDULA

Question:

Q. 4.13.105. Let

$$\overrightarrow{OP} = \frac{\alpha-1}{\alpha}\hat{i} + \hat{j} + \hat{k}, \quad \overrightarrow{OQ} = \hat{i} + \frac{\beta-1}{\beta}\hat{j} + \hat{k}, \quad \overrightarrow{OR} = \hat{i} + \hat{j} + \frac{1}{2}\hat{k}$$

be three vectors, where $\alpha, \beta \in \mathbb{R} - \{0\}$ and O denotes the origin. If

$$(\overrightarrow{OP} \times \overrightarrow{OQ}) \cdot \overrightarrow{OR} = 0$$

and the point $(\alpha, \beta, 2)$ lies on the plane

$$3x + 3y - z + l = 0,$$

then the value of l is _____.

(2024) **Solution:**

Solution:

Given

$$\mathbf{OP} = \begin{pmatrix} \frac{\alpha-1}{\alpha} \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{OQ} = \begin{pmatrix} 1 \\ \frac{\beta-1}{\beta} \\ 1 \end{pmatrix}, \quad \mathbf{OR} = \begin{pmatrix} 1 \\ 1 \\ \frac{1}{2} \end{pmatrix}. \quad (1)$$

From *gvvbook* (coplanarity/zero volume criterion), three vectors are coplanar iff the volume of the parallelepiped they form is zero, i.e.,

$$\det(\mathbf{OP} \ \mathbf{OQ} \ \mathbf{OR}) = 0. \quad (2)$$

Using (1),

$$\det \begin{pmatrix} \frac{\alpha-1}{\alpha} & 1 & 1 \\ 1 & \frac{\beta-1}{\beta} & 1 \\ 1 & 1 & \frac{1}{2} \end{pmatrix} = \frac{\alpha + \beta + 1}{2\alpha\beta} = 0 \implies \alpha + \beta + 1 = 0. \quad (3)$$

Let the plane be represented in normal form (cf. *gvvbook*, line/plane normal form)

$$\mathbf{n}^T \mathbf{x} + l = 0, \quad \mathbf{n} = \begin{pmatrix} 3 \\ 3 \\ -1 \end{pmatrix}. \quad (4)$$

Since the point $\begin{pmatrix} \alpha \\ \beta \\ 2 \end{pmatrix}$ lies on the plane,

$$\mathbf{n}^\top \begin{pmatrix} \alpha \\ \beta \\ 2 \end{pmatrix} + l = 0 \implies l = -(3\alpha + 3\beta - 2). \quad (5)$$

Using (3), $\beta = -1 - \alpha$, hence

$$l = -(3\alpha + 3(-1 - \alpha) - 2) = -(-5) = \boxed{5}. \quad (6)$$

3D Plot of Vectors OP, OQ, OR and Plane

